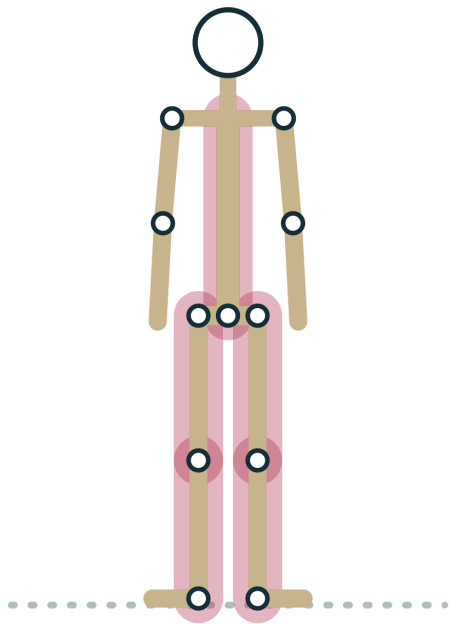


Module 2

The Skeleton for Yoga Teachers

Bones are the frame every pose is built on, and the landmarks you can see are how a teacher reads that frame in a room.



2 hours • 5 lessons • Dr. Akash Garg, D.C.
Anatomy & Physiology for Vinyasa Teachers

Overview

This module gives you a working map of the skeleton: not every one of the 206 bones, but the major ones you will name, feel and cue in every class. You will learn the difference between the axial skeleton, the protective central column of skull, spine and ribs, and the appendicular skeleton of limbs and girdles. Then you will follow the body from the skull down the spine, out through the shoulder to the weight-bearing hand, and down through the pelvis to the foot as the foundation of standing poses.

The final lesson turns anatomy into observation. Six bony landmarks, the ASIS, iliac crest, greater trochanter, acromion, C7 and malleoli, let you see alignment from across the room and help students find it in their own bodies. Throughout, we stay honest about what bones can and cannot do: bone shape differs between people, so range differs too, and a teacher's job is to adapt and refer, never to force or diagnose.

By the end of this module you can

1. By the end of this module you can classify the major bones as axial or appendicular and describe five functions of bone.
2. By the end of this module you can name the regions of the spine, their vertebra counts and their curves, and explain why twists come mainly from the thoracic spine.
3. By the end of this module you can identify the bones of the shoulder girdle, arm and hand, and give options for students with wrist discomfort in weight-bearing poses.
4. By the end of this module you can describe the pelvis, femur, lower leg and foot, including the foot tripod, and explain why hip range varies between people.
5. By the end of this module you can locate the ASIS, iliac crest, greater trochanter, acromion, C7 and malleoli on yourself and use them in clear, non-forcing cues.

Lessons

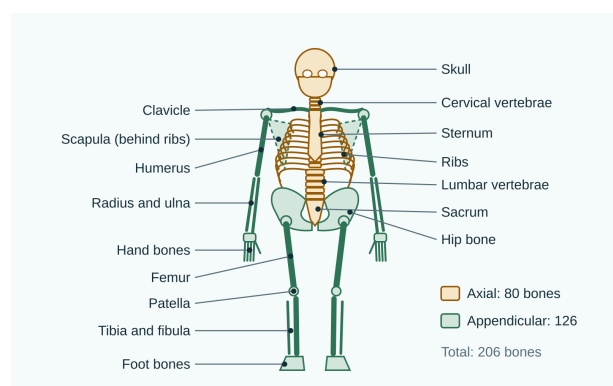
- **2.1** Axial and appendicular skeleton: what bones do (20 min)
- **2.2** Skull, spine, ribs and sternum (25 min)
- **2.3** Shoulder girdle and upper limb, down to the weight bearing hand (25 min)
- **2.4** Pelvis and lower limb, down to the foot as foundation (25 min)
- **2.5** Landmarks you can see and cue: ASIS, iliac crest, greater trochanter, acromion, C7, malleoli (25 min)

2.1 Axial and appendicular skeleton: what bones do

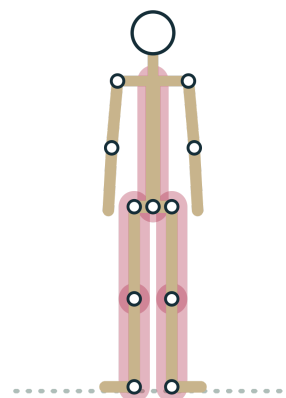
The adult skeleton has 206 bones. For teaching, the most useful split is between the axial skeleton and the appendicular skeleton. The axial skeleton, about 80 bones, runs down the midline: the skull, the vertebral column, the ribs and the sternum (breastbone). It protects the brain, spinal cord, heart and lungs, and it is the framework for posture and breathing. The appendicular skeleton, about 126 bones, is everything that hangs from that axis: the shoulder girdle (clavicles and scapulae), the arms and hands, the pelvic girdle (the two hip bones), and the legs and feet.

Bones have several jobs. They support the body, protect organs, and act as levers for muscles, which is how every pose happens. They also store calcium and phosphate, store fat in yellow marrow, and make blood cells in red marrow, found in places like the hip bones, sternum and vertebrae. Bones come in different shapes that suit their jobs: long bones such as the femur, short bones such as the wrist bones, flat bones such as the sternum and skull plates, irregular bones such as the vertebrae, and sesamoid bones such as the kneecap, which sits inside a tendon.

Bone is living tissue. It is constantly broken down and rebuilt, and it adapts to the loads it meets. Regular weight-bearing tends to help maintain bone, and lack of load weakens it. Yoga includes a lot of weight-bearing through both arms and legs, but research on yoga and bone density is limited and mixed, so teachers should not claim that yoga reverses osteoporosis. For students with known osteoporosis, teachers adapt: avoid forceful loaded spinal flexion and deep weighted twists, make balance safe, and defer to the student's medical team.



Axial skeleton in amber, appendicular in green



Mountain: bones stacked to carry load with little effort

KEY POINTS

- 206 adult bones: about 80 axial and 126 appendicular.
- Axial = skull, spine, ribs, sternum. Appendicular = limbs plus shoulder and pelvic girdles.
- Bones support, protect, act as levers, store minerals and fat, and make blood cells.
- Bone remodels throughout life and responds to load.
- Evidence that yoga builds bone is limited: adapt for osteoporosis and refer, never promise a cure.

FEEL IT

Stand in Mountain and tap your skull, breastbone and a few ribs to find your axial skeleton. Move into Tree and notice that the standing leg, an appendicular stack, carries the whole axial column. Raise your arms and feel your shoulder blades glide over the back of your ribs. Take three slow breaths and feel the ribcage widen and narrow.

CUE IT

“Let your bones stack so your muscles can soften a little.”

“Feel the standing leg carry the weight of your spine and head.”

“As you breathe in, let the ribs widen in every direction.”

CHECK IT

Which bone belongs to the appendicular skeleton?

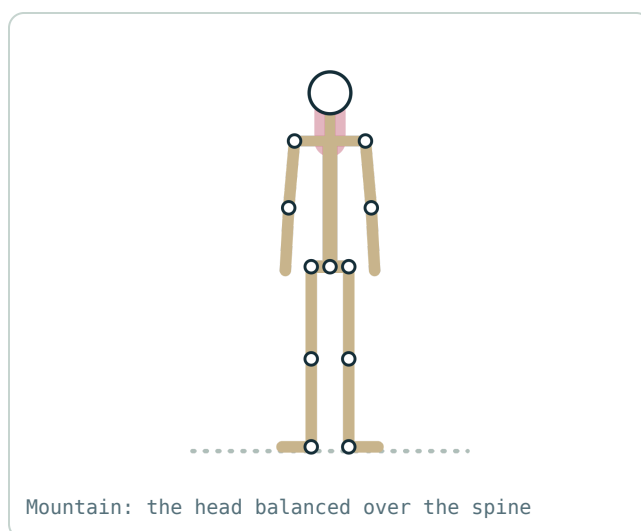
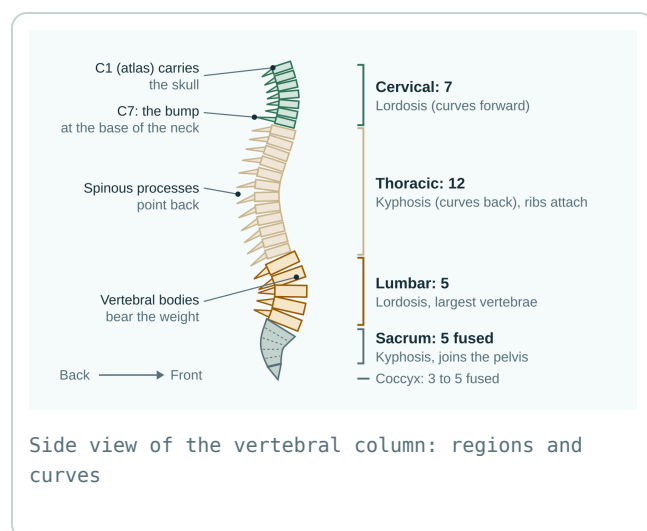
Answer: Scapula. The scapula is part of the shoulder girdle, which connects the arm to the axial skeleton.

2.2 Skull, spine, ribs and sternum

The skull sits on top of the axial skeleton. The cranium is a case of flat bones joined by fixed sutures, and the facial bones form the face. The mandible is the only freely moving skull bone. The occipital bone at the back of the skull rests on C1, the atlas, where much of nodding happens. C1 pivots on C2, the axis, and that pair provides about half of the head's rotation. The head weighs roughly 4.5 to 5 kg, which is why head position matters in every pose.

The vertebral column has 7 cervical, 12 thoracic and 5 lumbar vertebrae, then the sacrum (five fused) and the coccyx (three to five fused). Seen from the side it has four curves: cervical and lumbar lordosis (curving forward) and thoracic and sacral kyphosis (curving back). A typical vertebra has a weight-bearing body at the front, a disc between it and its neighbours, and an arch at the back protecting the spinal cord, with a spinous process you can feel as a bump under the skin. Each region moves differently. The cervical spine is the most mobile. The thoracic spine rotates well but its ribs limit backbending. The lumbar spine flexes and extends freely but rotates only a few degrees per segment, so twists should come mainly from the thoracic spine, neck and hips.

The thoracic cage is 12 pairs of ribs, the sternum and the thoracic vertebrae. Ribs 1 to 7 are true ribs attached to the sternum by their own cartilage, ribs 8 to 12 are false ribs, with 8 to 10 joined through the cartilage above and 11 and 12 floating free. The sternum has a manubrium, a body and a xiphoid process. Costal cartilage lets the cage expand as we breathe. When teaching twists, be modest: they move the spine and feel good, but the popular claim that they detox the organs is not supported by evidence.



KEY POINTS

- Skull rests on C1 (atlas); C1 turns on C2 (axis).
- 7 cervical, 12 thoracic, 5 lumbar, then sacrum and coccyx.
- Four curves: cervical and lumbar lordosis, thoracic and sacral kyphosis.
- Twist mostly from the thoracic spine; the lumbar spine rotates little.
- 12 rib pairs: 7 true and 5 false, of which 11 and 12 float; cartilage allows breathing movement.

FEEL IT

Stand with your heels, pelvis and upper back against a wall. Slide a hand behind your low back and then behind your neck to feel the two forward curves. Place your hands on your side ribs and breathe in slowly to feel them widen. Finally, trace your sternum from the notch at the top down to the xiphoid tip.

CUE IT

“Grow tall through the crown of the head, then turn from the ribs.”

“Let the back of the neck stay long, as if the head floats on top of the spine.”

“Breathe into the sides and back of your ribs.”

CHECK IT

A student forces a deep seated twist by levering with the arms. Which region is least built for rotation?

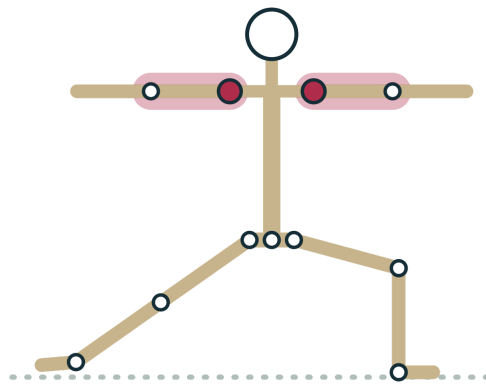
Answer: Lumbar spine. Lumbar vertebrae allow only a few degrees of rotation each, so forced twisting tends to load the low back.

2.3 Shoulder girdle and upper limb, down to the weight bearing hand

The shoulder girdle is made of the clavicle and scapula on each side. The clavicle joins the sternum at the sternoclavicular joint, the only bony connection between the arm and the trunk. The scapula glides over the back of the ribs, held there by muscle. Its outer corner has the acromion, which forms the bony top of the shoulder, and the glenoid fossa, a shallow socket that receives the head of the humerus. The shallow socket gives the shoulder the largest range of motion in the body but leaves it dependent on muscles for stability.

The humerus is the upper arm bone. At the elbow it meets the ulna on the little finger side and the radius on the thumb side. The olecranon of the ulna is the point of the elbow. The radius rotates around the ulna to turn the palm down (pronation) or up (supination), which is why a student can keep the palms flat in Downward Dog while rotating the upper arms outward.

Each wrist and hand has 27 bones: 8 carpals in two rows, 5 metacarpals and 14 phalanges. The carpals form an arch that makes the floor and walls of the carpal tunnel, where finger tendons and the median nerve pass. The hand is designed for grip and skill, yet vinyasa asks it to bear weight with the wrist extended close to its end range, roughly 90 degrees in Plank compared with a typical range of about 70 degrees. Teachers can help by cueing students to spread the fingers and press through the knuckles, especially the base of the index finger and thumb, so load is shared across the whole hand. Offer options such as a wedge under the heels of the hands, fists, or forearms, and refer on if pain, numbness or tingling persists.



Warrior II: arms carried by the shoulder girdle

KEY POINTS

- Shoulder girdle = clavicle + scapula; only bony link to the trunk is the sternoclavicular joint.
- Shallow glenoid socket: very mobile, needs muscular support.
- Humerus above; radius (thumb side) and ulna (little finger side) below.
- 27 bones per wrist and hand: 8 carpals, 5 metacarpals, 14 phalanges.
- Weight-bearing hands: spread load through knuckles and fingers, offer wrist options.

FEEL IT

Come to tabletop with your wrists roughly under your shoulders. Rock your weight into the heels of your hands, then into your fingertips, and notice how different they feel. Settle in the middle and press through the base of each finger, especially the index finger and thumb. Feel the centre of the palm lift slightly, like the arch of a foot.

CUE IT

“Press the base of your index finger and thumb down into the mat.”

“Let your shoulder blades glide wide across your back.”

“If your wrists complain, come onto your fists or forearms.”

CHECK IT

Which is the only joint that connects the upper limb bones to the axial skeleton?

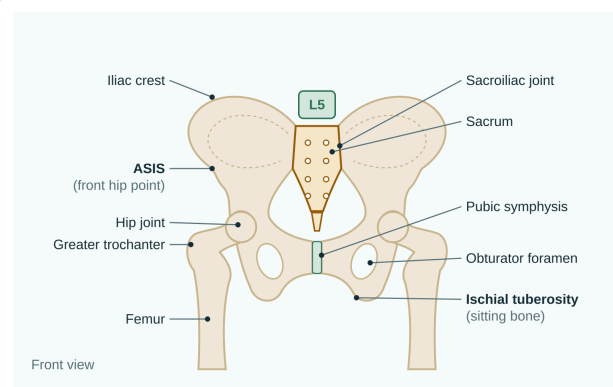
Answer: Sternoclavicular joint. The clavicle meets the sternum at the sternoclavicular joint; everything else attaching the arm to the trunk is muscle.

2.4 Pelvis and lower limb, down to the foot as foundation

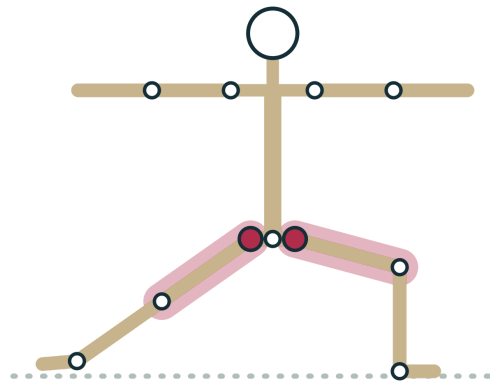
The bony pelvis is a ring made of two hip bones and the sacrum, with the coccyx below. Each hip bone forms from three bones that fuse in the teenage years: the ilium (the broad wing, topped by the iliac crest and ending in front at the ASIS), the ischium (whose ischial tuberosity is the sitting bone) and the pubis (which meets its partner at the pubic symphysis). At the back, each hip bone joins the sacrum at a strong sacroiliac joint that moves only a few degrees. The three parts meet at the acetabulum, the deep socket for the femoral head. The hip bones form the pelvic girdle and are appendicular, while the sacrum and coccyx belong to the axial spine.

Compared with the shoulder girdle, the pelvic girdle is heavier, closed and firmly joined to the spine, with a deep socket. It is built for stability and weight bearing rather than range. The femur is the longest, strongest bone. Its head sits in the acetabulum, its neck angles outward to the greater trochanter, and its shaft slopes inward to the knee. The angles of the femoral neck and the socket vary between people, so hip range in poses like Warrior II, Pigeon or Lotus differs partly because of bone shape, not only muscle length. Teachers should never force a hip past its end point.

At the knee the femur rests on the tibia, and the patella, a sesamoid bone inside the quadriceps tendon, glides in front. The tibia carries most of the body's weight; the slender fibula mainly anchors muscles. Their lower ends form the medial and lateral malleoli. Each foot has 26 bones: 7 tarsals, 5 metatarsals and 14 phalanges, arranged in medial, lateral and transverse arches. A helpful teaching model is the foot tripod: the big toe mound, little toe mound and heel. Arch height varies widely in healthy people, so teachers invite even grounding rather than diagnosing flat feet.



Pelvis from the front, with the key landmarks



KEY POINTS

- Hip bone = ilium + ischium + pubis, fused; pelvis = two hip bones + sacrum (+ coccyx).
- Acetabulum is a deep socket: the hip trades range for stability.
- Femur: head, angled neck, greater trochanter; bone shape varies between people.
- Tibia carries weight; fibula anchors muscles; malleoli are their lower ends.
- 26 bones per foot, three arches, and a three-point tripod for grounding.

FEEL IT

Stand in Mountain and lift all ten toes. Notice the weight settle into three points: the big toe mound, the little toe mound and the centre of the heel. Keep those points grounded as you lower the toes softly, and notice whether the inner arch lifts slightly. Rock gently forward and back to find the place where all three points share the load.

CUE IT

“Root down through the big toe mound, the little toe mound and the heel.”

“Let your knees point the same way as your second and third toes.”

“Sit down onto your sitting bones and let the spine rise up from them.”

CHECK IT

Which bone forms the outer ankle bone, the lateral malleolus?

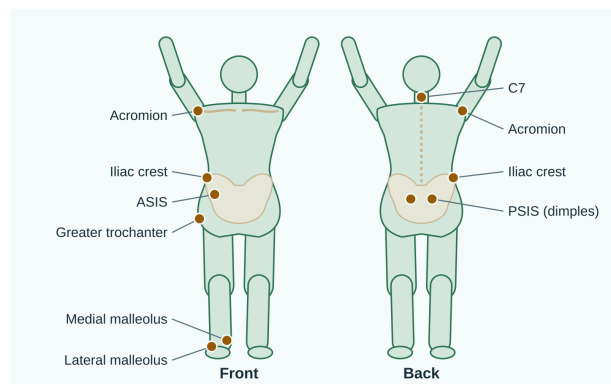
Answer: Fibula. The lateral malleolus is the lower end of the fibula; the medial malleolus is the lower end of the tibia.

2.5 Landmarks you can see and cue: ASIS, iliac crest, greater trochanter, acromion, C7, malleoli

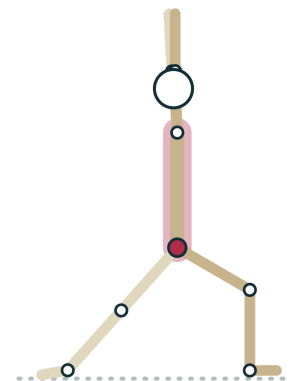
Bony landmarks are points of bone near the surface that you can see or feel. They make anatomy visible in a class and give students something concrete to find on themselves. Six are especially useful. The ASIS (anterior superior iliac spine) is the front hip point at the forward end of the iliac crest. It shows which way the pelvis faces, for example in Warrior I, where many students cannot square the hips fully because of bone shape and limited back hip extension. The iliac crest is the curved top of the pelvis where hands rest on hips; a line across the tops of both crests crosses the spine at about L4. Just below the back of the crests are the PSIS dimples beside the sacrum.

The greater trochanter is the bony bump on the outer upper thigh, part of the femur. It rolls forward and back as the thigh rotates, which helps students feel hip rotation in Warrior II or Triangle. The acromion is the flat bony shelf at the top of the shoulder, at the outer end of the scapula; it shows whether the shoulders are lifted toward the ears. C7 usually has the most prominent spinous process at the base of the neck. If the bump you touch slides forward when you look up, it is C6, and C7 is the next one down. The medial malleolus (end of the tibia) is the inner ankle bone and the lateral malleolus (end of the fibula) is the outer one, sitting lower and further back. In Tree, the lifted foot presses above or below the knee, not on it.

Use landmarks to observe and to cue self-discovery. Bodies are rarely perfectly symmetrical, so an uneven crest or shoulder is something to notice, not to diagnose. Invite students to touch their own landmarks, and ask for clear consent before any hands-on assist. When something looks unusual or a student reports pain, adapt the pose and suggest they speak to a health professional.



Six key landmarks from the front and back



Warrior I: the ASIS points show where the pelvis faces

KEY POINTS

- ASIS = front hip points; they show where the pelvis faces.
- Iliac crests = hands-on-hips line, level with about L4; PSIS dimples sit below at the back.
- Greater trochanter rolls as the thigh rotates.
- Acromion = top of the shoulder; C7 = biggest bump at the base of the neck.
- Medial malleolus (tibia) is higher; lateral malleolus (fibula) is lower. Observe, never diagnose.

FEEL IT

Stand in Mountain with your hands on your hips to find the iliac crests, then slide forward to the ASIS points and back to the PSIS dimples. Place a hand on the greater trochanter and turn your foot in and out to feel it roll. Trace your collarbone out to the acromion, find the biggest bump at the base of your neck, and look up to check it is C7. Finish by touching your inner and outer ankle bones.

CUE IT

“Bring your front hip points toward the front of the mat, only as far as feels good.”

“Let the tops of your shoulders slide away from your ears.”

“Press your foot into your inner thigh or calf, above or below the knee.”

CHECK IT

In Warrior I a student's back hip will not turn to face forward. What is the best response?

Answer: Invite the hips toward the front as far as comfortable, and offer a shorter, wider stance. Bone shape and back hip range limit how square the pelvis can get, so invite and offer options rather than force or diagnose.

Key terms

Axial skeleton	The central column of about 80 bones: skull, spine, ribs and sternum. In Mountain it is the stacked core the limbs hang from.
Appendicular skeleton	The 126 bones of the limbs and the shoulder and pelvic girdles. In Tree the standing leg is appendicular.
Vertebra	One of the bones of the spine, with a weight-bearing body in front and a protective arch behind. You feel their spinous processes as bumps when rolling down.
Lordosis	A forward curve of the spine, found in the neck and low back. It is the hollow you feel behind your low back in Mountain against a wall.
Kyphosis	A backward curve of the spine, found in the thoracic and sacral regions. It is the gentle rounding of the upper back that touches the wall.
Intervertebral disc	A pad of cartilage between two vertebral bodies with a tough outer ring and a gel-like centre. It shares load as you bend in Forward Fold.
Atlas and axis (C1, C2)	The top two cervical vertebrae. The skull nods on the atlas and turns as the atlas pivots on the axis, as when you look over your shoulder in Warrior II.
Sternum	The breastbone, made of the manubrium, body and xiphoid process. It lifts gently as the chest opens in Cobra.
Clavicle	The collarbone, joining the sternum to the scapula. It lifts slightly when you reach the arms overhead in Warrior I.
Scapula	The shoulder blade, which glides over the back of the ribs. It slides wide across the back in Plank.
Radius and ulna	The two forearm bones: radius on the thumb side, ulna on the little finger side. The radius crosses the ulna when the palms press flat in Down Dog.
Carpals	The eight small bones of the wrist in two rows. They carry weight through the heel of the hand in Plank.
Ilium, ischium and pubis	The three bones that fuse to form each hip bone. You sit on the ischium's sitting bones in Easy Pose.
Acetabulum	The deep cup-shaped hip socket that holds the head of the femur. The femur rotates within it in Warrior II.
Femur	The thighbone, the longest and strongest bone. Its angled neck and head shape help set how far a student can go in Lotus.
Patella	The kneecap, a sesamoid bone inside the quadriceps tendon. It lifts as you engage the thighs in Mountain.
Tibia and fibula	The lower leg bones: the tibia carries most of the weight, the slender fibula mostly anchors muscles. The tibia stacks over the ankle in Chair.

Foot tripod	The three grounding points of the foot: big toe mound, little toe mound and heel. Rooting all three steadies Tree.
ASIS	Anterior superior iliac spine: the front hip point at the end of the iliac crest. Its direction shows where the pelvis faces in Warrior I.
Greater trochanter	The bony bump on the outer upper thigh, part of the femur. It rolls back as the thigh turns out in Warrior II.
Acromion	The flat bony shelf at the top of the shoulder, part of the scapula. Cue it away from the ears in Warrior II.
Malleoli	The ankle bones: medial (inner, from the tibia) and lateral (outer, from the fibula). In Tree the foot rests above or below the knee, not on the knee joint.

Myths to retire

Bones are dead, fixed structures.

Bone is living tissue that is constantly broken down and rebuilt and responds to the loads placed on it.

Yoga reverses osteoporosis.

Weight-bearing helps bone in general, but evidence that yoga improves bone density is limited and mixed. Adapt poses and follow the student's medical advice.

Everyone can square the hips in Warrior I with enough effort.

Hip range depends partly on bone shape and back hip extension. Invite the hips toward the front as far as comfortable instead of forcing.

Twists wring out toxins from the organs.

There is no good evidence for detox by twisting. Twists move the spine, mostly at the thoracic level, and can simply feel good.

Flat feet or an uneven pelvis mean something is wrong.

Arch height and minor asymmetry vary widely in healthy people. Teachers observe and adapt; they do not diagnose.

Self check

1. What is the difference between the axial and appendicular skeleton?
2. Name the four spinal curves and whether each is lordotic or kyphotic.
3. Why should a twist come mainly from the thoracic spine?
4. Why is the shoulder more mobile but less stable than the hip?
5. Give two options for a student with wrist pain in Downward Dog.
6. What is the foot tripod and why is it useful?
7. How would you use the ASIS in Warrior I without forcing?
8. Which bones form the medial and lateral malleoli, and which sits lower?

Model answers

1. The axial skeleton is the central column of skull, spine, ribs and sternum, about 80 bones. The appendicular skeleton is the limbs and the shoulder and pelvic girdles, about 126 bones.
2. Cervical and lumbar curves are lordotic, curving forward. Thoracic and sacral curves are kyphotic, curving backward.
3. Thoracic joints allow good rotation while the lumbar spine rotates only a few degrees per segment. Forcing a twist loads the low back.
4. The glenoid socket is shallow and the shoulder girdle floats on the ribs, attached by bone only at the sternoclavicular joint. The hip has a deep socket in a pelvic ring joined to the spine.
5. Cue them to spread the fingers and press through the knuckles, and offer a wedge under the heels of the hands, fists or forearms. Refer on if pain or tingling persists.
6. The big toe mound, little toe mound and heel. Grounding all three spreads load and steadies balance poses like Tree.
7. Invite the front hip points toward the front of the mat only as far as comfortable, and offer a shorter, wider stance if the back hip cannot turn.
8. The medial malleolus is the end of the tibia and the lateral malleolus is the end of the fibula. The lateral one sits lower.